

KalaSetu AI (कलासेतु)

A Sovereign, AI-Powered Rural Artisan Market Linkage, Conscious Heritage Commerce & Financial Inclusion Ecosystem.

✦ Core Ecosystem Philosophy:

Discover → Understand the Story → Experience the Craft → Support the Artisan → Purchase → Continue Supporting

TOTAL FEATURES DOCUMENTED
25 Master Capabilities

TARGET BENEFICIARIES
7M+ Rural Handloom & Craft Artisans

AI RUNTIME MODEL
100% Free / Open Source (RTX CUDA)

CLIENT FOOTPRINT
< 150 MB RAM (Android APK & PWA)

Section 1: The Real Problem & Market Loopholes KalaSetu Solves

ROOT CAUSE ANALYSIS

1. The "Retailer-Connection" Trap

Mainstream e-commerce platforms (Amazon Karigar, Etsy, Flipkart) link customers with urban aggregators and retail wholesalers. The actual village creator gets zero visibility, zero brand recognition, and only 25%–30% of final retail value.

KalaSetu's Direct Artisan Attribution

Zero-middleman direct linkage. Every product features an immutable digital **Artisan Pehchan Card**, direct QR contact, and personal audio lineage. Buyers connect directly with the maker.

2. The Digital Literacy & Form Barrier

Conventional seller portals require typing, complex SKU categorization, GST details, English descriptions, and high-spec desktop setups—impossible for rural artisans.

Voice-First Vernacular Cataloging

The artisan speaks in their local dialect for 10 seconds: "बगरू हैंड-ब्लॉक दुपट्टा, प्राकृतिक नील रंग, 2 दिन लगे, 1200 रुपये". KalaSetu's AI extracts, structures, and creates luxury listings instantly.

3. Counterfeits & Unverifiable Claims

Industrial powerlooms and factory machine-prints sell cheap polyester as "Pure Handwoven Silk" or "Natural Indigo", destroying the market value of authentic handmade crafts.

ClaimSafe AI & Verifiable Product Passport

AI scans 16+ high-risk claims, enforces hard publishing gates, and issues a tamper-evident Product Passport verifying craft technique, materials, and GI heritage cluster.

4. The Financial Blindspot (Unclaimed Schemes)

Over ₹13,000 Cr in government support (PM Vishwakarma, Pehchan IDs, Mudra loans) remains unclaimed by rural artisans due to lack of awareness and complex paperwork.

Kala Funds Financial Aid Hub

An integrated scheme engine where artisans discover eligible welfare programs, toolkits, collateral-free credit, and NGO grants in 1 tap directly from their phone.



Section 2: Complete Master Feature Matrix (All 25 Capabilities)

100% Comprehensive

PILLAR 1: CONSCIOUS HERITAGE COMMERCE & BUYER DISCOVERY

1. Premium Artisan Marketplace

Curated high-value positioning with luxury tiering. Moves beyond generic handicrafts to showcase museum-grade heirloom masterworks.

2. Emotional Artisan Backstories

Every product highlights the artisan's personal lineage, multi-generational heritage, and the emotional human narrative behind its creation.

3. Expanded Product Categories

Comprehensive categories: Tribal Jewellery, Leather Mojari, Brass Dhokra, Terracotta, Textiles, Handloom Silks, and Wood Carvings.

4. Kala Reels → Direct Purchase

Short-form videos of artisans at work (spinning yarn, firing kilns). Integrated floating 1-tap checkout: *Discover* → *Watch* → *Buy*.

5. AI Room Décor Simulator

Space Stager: Buyers upload a photo of their wall, table, or room; AI visualizes pottery, rugs, and wall hangings placed in their real space.

6. AI Multi-Object Visual Search

Upload 1 lifestyle photo to detect multiple objects simultaneously (e.g., [Necklace] + [Silk Scarf]) and match each to authentic artisans.

7. AI Suggestions & Stock Alternatives

Recommends complementary artisan crafts. When a unique piece is sold out, automatically directs buyers to similar crafts by peer artisans.

8. AI Curated Digital Gallery

Virtual artisan exhibition halls where creators showcase cohesive thematic collections, enhancing prestige and global reach.

9. Direct Artisan Link (No Middlemen)

Disintermediates aggregators and retailers. Direct chat, custom commissioning, and transparent maker attribution.

PILLAR 2: SMART CREATION, CREATION INTEGRITY & WAGE DEFENSE

10. Voice-to-Catalog PWA

14+ regional dialects supported. 10-second vernacular voice description generates structured title, specs, materials, and tags.

11. ClaimSafe AI Verification

Scans 16+ high-risk claims (*Pure silk, Natural dye, GI registered*). Hard validation gate prevents fraudulent consumer claims.

12. FairPrice Guard & Anti-Exploitation

Cost-plus formula: $\text{Materials} + (\text{Hours} \times \text{Rate}) + \text{Overhead}$. Warns against ₹0 labor and assists in buyer counter-negotiation.

13. Verifiable Product Passport

Buyer-facing digital authenticity certificate with QR code, verified materials, cluster mapping, and strict privacy consent defaults.

14. Truthful Image Studio

Non-destructive canvas lighting adjustment with strict anti-synthetic policy, preserving real handmade imperfections.

15. Craft-Aware Missing Field Wizard

Contextual templates for Bagru Prints, Blue Pottery, Dhokra, etc. Single-question wizard prompts for crucial missing provenance details.

16. Offline Drafts & Conflict Queue

IndexedDB v2 dual stores with 5 sync states. Operates 100% offline in rural villages; creates safe copies during sync conflicts.

17. Hisab-Kitab Daily Business Ledger

Camera OCR scans paper expense slips and bills; tracks daily raw material expenses, sales revenue, and voice-assisted profit summaries.

18. Artisan Productivity Score

Comprehensive 0–100 business health score tracking fulfilled orders, catalog updates, hisab consistency, and low returns.

PILLAR 3: FINANCIAL AID, GOVERNANCE & COMMUNITY ECOSYSTEM

19. Kala Funds (Financial Aid Hub)

20. Pehchan Artisan ID Integration

21. Governance & Institutional Links

Discover & apply for government schemes: PM Vishwakarma, AHVY, Mudra loans, and NGO grants in 1 click directly in the app.

Embeds national artisan card credentials into the Product Passport for free accidental insurance and government haat quota verification.

Partnerships with organizations like Human Planet, TRIFED, and ONDC for ethical certification, transparency, and global fair trade.

22. Facilitator & NGO Audit Desk

Dedicated 5-metric dashboard for cluster coordinators and SHG leads to audit claims, verify prices, and track artisan community growth.

23. Cluster Geo-Fencing Dispatch

Validates that product dispatch origin physically matches the recognized geographical heritage boundary (e.g. Bagru cluster polygon).

24. KalaSetu Saathi (Voice AI)

Conversational assistant for buyers (heritage gift finder) and artisans (daily profit, stock alerts, voice-guided help) in native tongues.

25. Support Hubs & Rewards

Raw material bulk buying circles, artisan legal/design mentorship, Artisan Karma badges, and Patron rewards for repeat conscious buyers.



Section 3: Technical System Architecture & Hardware Specs

Zero-Billing Sovereign AI

KalaSetu AI solves the cost barrier of generative AI by running state-of-the-art open models on consumer gaming hardware (RTX 3050 CUDA) bridged to a featherlight mobile APK via Cloudflare Tunnel.



SYSTEM LAYER	TECHNOLOGY	HOST ROLE	RESOURCE FOOTPRINT
Mobile Client	Android TWA / PWA (PWABuilder)	Artisan UI, Mic, Camera, Offline DB	< 150 MB RAM
Language Model	Qwen2.5-3B-Instruct (Q4_K_M)	Catalog structuring & negotiation reasoning	2.0 GB VRAM (RTX 3050)
Vision Model	Moondream2 (1.86B Q4)	Motif identification & craft photo inspection	1.4 GB VRAM (RTX 3050)
Document OCR	Tesseract.js (v5) + Devanagari	Handwritten expense slips & Pehchan ID OCR	80 MB Client WASM
Cloud Bridge	Cloudflare Tunnel (cloudflared)	Zero-config HTTPS tunnel to mobile APK	15 MB System RAM



Section 4: Kala Funds — Government Schemes & Financial Inclusion Hub

Empowerment Engine

SCHEME / PROGRAM	TARGET BENEFICIARY	KEY FINANCIAL / MATERIAL BENEFIT	KALASETU INTEGRATION
PM Vishwakarma Yojana	18 traditional artisan trades (Potters, Weavers, Blacksmiths)	₹15,000 tool kit incentive + ₹1 Lakh (Tranche 1) & ₹2 Lakh (Tranche 2) @ 5% concessional interest.	1-Click eligibility check, craft certification pre-fill, and application tracking.
Pehchan Artisan ID Card	All verified handicraft & handloom artisans	Official national artisan identity, free accidental insurance coverage (₹2 Lakh), and direct haat stalls.	Integrated directly into the Product Passport QR code as proof of master craft lineage.
Ambedkar Hastshilp Vikas (AHVY)	Artisan Self-Help Groups (SHGs) and rural craft clusters	Common Facility Centers (CFC), raw material bank subsidies, and modern design workshops.	Facilitator/NGO desk clusters artisans into compliant SHGs to claim bulk infrastructure grants.
Pradhan Mantri Mudra Yojana	Micro-enterprises & individual artisans	Shishu (up to ₹50,000) & Kishore (up to ₹5 Lakh) collateral-free working capital loans.	Artisan Productivity Score serves as an alternative non-traditional creditworthiness benchmark.



Section 5: Measurable Social Impact & Market Viability

ECONOMIC MULTIPLIER

Artisan Income Multiplier (+140% Real Take-Home)

By eliminating the 4-layer middleman chain (Village broker → City wholesaler → Metro retailer → Export agent), direct-to-consumer provenance sales boost the artisan's take-home earnings from ₹350 to ₹850 on an authentic Bagru dupatta.

Zero Recurring Cloud Costs for NGOs & Self-Help Groups

Because the entire AI pipeline executes locally on consumer-grade gaming hardware (RTX 3050 CUDA) or in-browser WASM, rural cooperatives can operate the platform indefinitely with ₹0 monthly API subscriptions.